

A compact-operator model for a representation

MathlibAnnex.Analysis.CStarAlgebra.IsCompactOperatorModel

def

Packages faithfulness together with the equality $e(A) = \mathcal{K}(H)$.

Statement

Let A be a complex C^* -algebra, with no unit assumed, let H be a complex Hilbert space, and let $e : A \rightarrow \mathcal{B}(H)$ be a $*$ -representation. The predicate below asserts that e identifies A exactly with the compact operators on H .

Definition

The three conditions are

e is injective, $e(A) \subseteq \mathcal{K}(H)$, $\mathcal{K}(H) \subseteq e(A)$.

Equivalently, e is faithful and $e(A) = \mathcal{K}(H)$; hence A is $*$ -isomorphic to $\mathcal{K}(H)$ through the specified representation.

Assumptions

No separability, irreducibility or singleton-spectrum hypothesis is part of this definition. A unit is not assumed.

Conclusion

The output is a property of the given representation e , not merely an assertion that some abstract $*$ -isomorphism exists.

Main citations

- Exact definition and context

Lean source signature (exact)

The complete declaration below is a separate exact source excerpt; the original header record is retained with the manuscript.

```
/-- A nonunital star representation that is injective and has precisely the
compact operators as its range. This is the unbundled form of a star
isomorphism with `K(H)`. -/
def IsCompactOperatorModel [NonUnitalCStarAlgebra A]
  (e : A →*na [C] (H →L[C] H)) : Prop :=
  Function.Injective e ∧
```

$$(\forall a : A, \text{IsCompactOperator } (e\ a)) \wedge \\ \forall T : H \rightarrow L[\mathbb{C}]\ H, \text{IsCompactOperator } T \rightarrow \exists a : A, e\ a = T$$

In the source	Mathematical meaning
<code>[NonUnitalCStarAlgebra A]</code>	The complex C^* -algebra A ; a unit is not required.
<code>(e : A →*_{na} [C] (H →L [C] H))</code>	The given $*$ -representation $e : A \rightarrow \mathcal{B}(H)$.
<code>Function.Injective e</code>	Equal represented operators imply equal algebra elements.
<code>(∀ a : A, IsCompactOperator (e a))</code>	Every element a of A is sent to a compact operator; this is $e(A) \subseteq \mathcal{K}(H)$.
<code>∀ T : H →L [C] H, IsCompactOperator T → ∃ a : A, e a = T</code>	Choose any bounded operator T ; if it is compact, choose an element a of the original algebra whose image is exactly T . This is the opposite inclusion.
<code>: Prop</code>	The conjunction of injectivity and the two inclusions $e(A) \subseteq \mathcal{K}(H)$ and $\mathcal{K}(H) \subseteq e(A)$.

Exact source and proof.

Earlier published Card and PDF

▼ Exact content identity

Declaration: `MathlibAnnex.Analysis.CStarAlgebra.IsCompactOperatorModel`
 Accepted content SHA-256: `2ccd05514cd6571908decd8a6e1d8a3991d3eeb9e9dcdf01b7070510bdeaeaa8`
 Accepted source guide SHA-256: `d640d0b8566257ea8525354777d4a81352e71a27795c169ca2acfe7a7c614066`
 Source commit: `437e6e46228bbb8e91211ebded349d7a30020e73`